Introduction to Access

Outline

- *Understanding a database system*
- *Types of databases*
- *Understanding and creating Fields, Data Types, Field size, Records and other features in MS Access.*
- *Understanding the importance of Primary keys in relational databases*
- *Understanding and creating tables, forms, and reports*
- *Creating queries and calculated fields.*
- *Conclusion*
- *Reference*

"The function of education is to teach one to think intensively and to think critically. Intelligence plus character - that is the goal of true education".

Martin Luther King, Jr.

"The only source of knowledge is experience".

Albert Einstein.

What is a database?

This is a collection of related data that has been entered into a computer system and stored for future use.

A database is a collection of information organized in a certain order to provide efficient retrieval.

What is a DBMS?

A database Management System (DBMS) is a software package used to set up, or structure a database.

Examples of databases

- Microsoft Access
- Oracle
- FileMaker Pro
- Microsoft SQL Server
- SAP and MySQL

Types of databases

There are two types of databases namely:

- Flat database (single)
- Relational database (linked)

Flat File databases: These are databases that contain a single table of data.

A **relational database** is one that contains multiple tables of data that relate to each other through special key fields

Database objects

- A **table** is a data structure used to organize information into rows and columns. The database table is where all the data in a database is stored.
- A **form** is a **database** object that you can use to create a user interface for a **database** application.

Database Objects

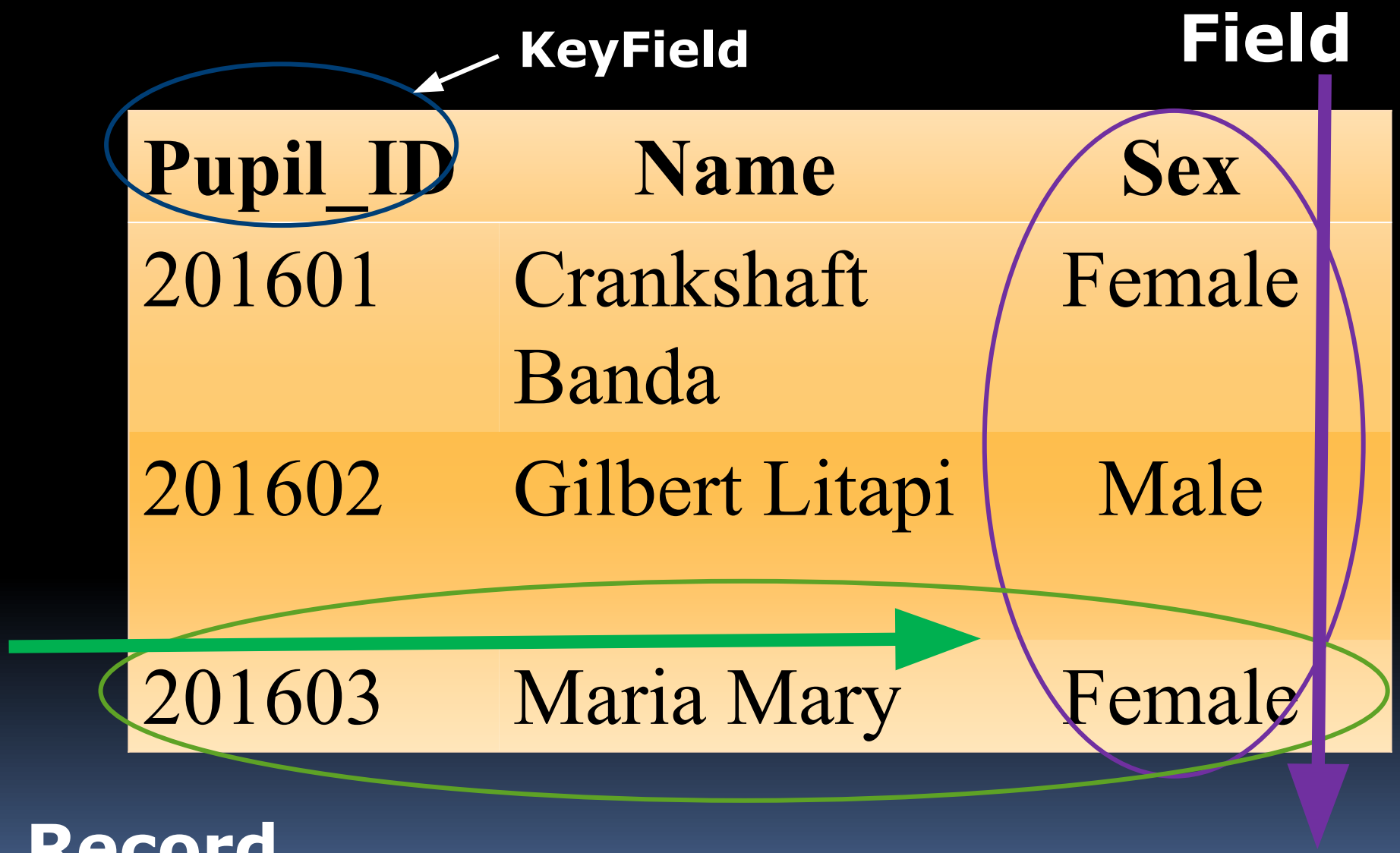
- A **database report** presents information that can be viewed, displayed and printed from the database.
- A **query** is a request for data or information from a **database** table or combination of tables. A **query** is used to retrieve data from the **database**

Database Elements

Databases are organised by fields, records and files.

- **Field** (Columns)- This is one piece of data or information about a person or thing e.g name, date of birth or telephone number.
- **Record** (Rolls)- this is one complete set of fields, for example one pupil's school record
- **File** - this is a collection of records, for example a school database

Database Elements



Database Elements

- A **Key field** is a field in a record that holds unique data which identifies that record from all the other records in the file or database.

Types of key fields:

- Primary Key, Secondary Key, Foreign key, Simple key, Compound key and Composite key

Database Elements

- A **primary key** is data that is unique to each record in a database or file. This prevents any records from having the same value.
- A secondary key is a non-unique data that is used to sort or improve record retrieval.

Database Elements

- A **foreign key** in a **database** table is a key from another table that refers to (or targets) a specific key usually the primary key.

OR

- A foreign key is used to link tables together and create a relationship. It is a field in one table that is linked to the primary key in another table.

Importance of key fields in a database

Keys fields are crucial to a table structure for the following reasons:

- They ensure that each record in a table is precisely identified or uniquely identified.
- They help establish and **enforce** various types of integrity.
- They serve to **establish table relationships**.

What is a relationship in a database?

A relationship exists between two database tables when one table has a foreign key that references the primary key of another table.

Types of relationships

- **One-to-one:** One to one is implemented using single table by establishing relationship between same type of columns in a table.
- **One-to-many:** Implemented using two tables with primary key and foreign key relationships.
- **Many-to-many:** Implemented using a junction table. The keys from both the tables form composite primary key of the junction table.

Explanation on relationships

- **One to One:** One entity is associated with another entity. E.g. Each employee is associated with one department.
- **One to Many:** One entity is associated with many other entities. E.g. A company is associated with all working employees in one branch/office/country.

Explanation on relationships

- **Many to One:** Many entities are associated with only one entity. E.g. Many employees are associated with one project.
- **Many to Many:** Many entities are associated with many other entities. E.g. In a company many employees are associated with multiple projects.

Data Type

- A data type is a type of data.
- Data type is a data storage format that can contain a specific type or range of values.
- A data type in a programming language is a set of data with values having predefined characteristics.

Examples of Data Types

- Integers: (Whole numbers)
- Characters (Char)/Text
- Floating point (Number with decimal points)
- Strings:(is a series of characters treated as a single unit)
- Arrays: (data structure that contains a group of elements.)
- Memo
- Alphanumeric (Alphabet and numbers)
- Text

Conclusion

A **database** management system is **important** because it manages data efficiently and allows users to perform multiple tasks with ease.

A **database** management system stores, organizes and manages a large amount of information within a single software application

Programming with MS Access

Hands on practice

Reference

- http://techterms.com/definition/array
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- Teach ICT: http://www.teach-ict.com/as_a2_ict_new/ocr/AS_G061/315_database_concepts/terminology/miniweb/pg12.htm
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